

Introduction: Rare Bleeding Disorders: General Aspects of Clinical Features, Diagnosis, and Management

Flora Peyvandi, M.D., Ph.D.,¹ Roberta Palla, Ph.D.,¹ Marzia Menegatti, Ph.D.,¹ and Pier Mannuccio Mannucci, M.D.¹

ABSTRACT

Rare bleeding disorders (RBDs) are autosomal recessive diseases including the inherited deficiencies of coagulation factors such as fibrinogen, factor (F) II, FV, FV + FVIII, FVII, FX, FXI, FXIII, and multiple deficiency of vitamin K-dependent factors, with clinical manifestations ranging from mild to severe. They represent 3 to 5% of all the inherited coagulation deficiencies with a prevalence in the general population varying between 1 in 500,000 and 1 in 2 million, being higher in areas where consanguineous marriages are diffuse. Despite the progress made in past years, as a consequence of the rarity of these deficiencies, the type and severity of bleeding symptoms, the underlying molecular defects, the actual management of bleeding episodes and particularly the prophylactic treatment in patients affected with RBDs are not well established. In this introductory article, the main features, diagnosis, available treatment options, and treatment complications of RBDs will be discussed.

KEYWORDS: Coagulation disorders, bleeding symptoms, phenotypic diagnosis, genotypic diagnosis, replacement therapy, rare bleeding disorders, rare coagulation disorders

The rare bleeding disorders (RBDs) include the inherited deficiencies of such coagulation factors as fibrinogen, factor (F) II, FV, FV + FVIII, FVII, FX, FXI, FXIII, and of multiple deficiency of vitamin K-dependent factors and are usually transmitted in an autosomal recessive manner and, due to their rarity, sometimes present significant difficulties in diagnosis and treatment.^{1,2} The limited available data makes it difficult for us to draw an exact picture of the prevalence of each RBD. Until 1980, only sparse information on

RBDs was available, whereas in the past decade, the number of reported studies have increased considerably, particularly regarding molecular aspects of RBDs. The prevalence of RBDs in the general population is low, and homozygous or double heterozygous deficiencies vary from 1:500,000 for FVII deficiency to 1:2,000,000 for prothrombin and FXIII deficiencies.¹⁻³ The prevalence of RBDs is also strongly influenced by the ethnicity mix and is significantly increased by a high rate of consanguinity in the population.³ The rarity of RBDs also

¹Angelo Bianchi Bonomi Hemophilia and Thrombosis Center, and Department of Medicine and Medical Specialties, IRCCS Maggiore Hospital, Mangiagalli and Regina Elena Foundation, University of Milan, Luigi Villa Foundation, Milan, Italy.

Address for correspondence and reprint requests: Flora Peyvandi, M.D., Ph.D., Associate Professor of Internal Medicine, A. Bianchi Bonomi Hemophilia and Thrombosis Center, Department of Medicine and Medical Specialties, IRCCS Maggiore Hospital, Mangiagalli and Regina Elena Foundation, University of Milan and Luigi Villa

Foundation, Via Pace 9, 20122 Milan, Italy (e-mail: flora.peyvandi@unimi.it).

Rare Bleeding Disorders; Guest Editor, Flora Peyvandi, M.D., Ph.D.

Semin Thromb Hemost 2009;35:349-355. Copyright © 2009 by Thieme Medical Publishers, Inc., 333 Seventh Avenue, New York, NY 10001, USA. Tel: +1(212) 584-4662.

DOI 10.1055/s-0029-1225757. ISSN 0094-6176.

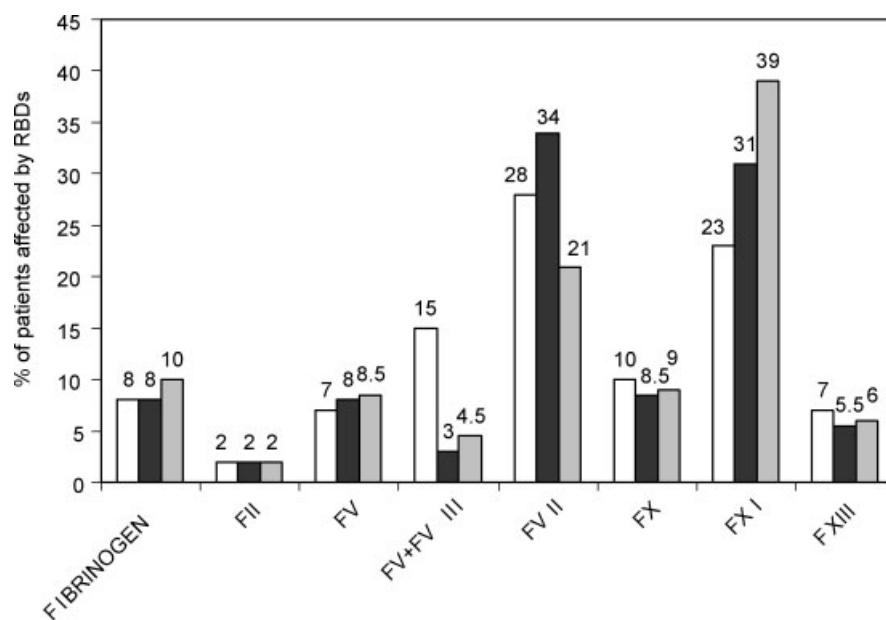


Figure 1 Worldwide prevalence of rare bleeding disorders derived from RBDD (white-filled bars), WFH (black-filled bars), and the cumulative results of six national registries (Central and South America, Egypt, France, India, Iran, United Kingdom; gray-filled bars).

influences diagnosis, monitoring, and treatment of affected individuals, which require the involvement of expert laboratories with specialized phenotypic and genotype facilities for an accurate diagnosis.⁴ The worldwide prevalence of RBDs can be derived from two large data collections, the first by the World Federation of Haemophilia (WFH; http://www.wfh.org/2/7/7_0_Link7_GlobalSurvey2005.htm) and the second by the International Rare Bleeding Disorders Database (RBDD; www.rbdd.org). The WFH started collecting information on hemophilia care throughout the world in 1998, but collection of data on RBDs started in 2004. Its current global survey includes basic demographic information, data on care resources and treatment products, and information on the prevalence of infectious complications such as HIV and hepatitis C in people with hemophilia, von Willebrand disease, platelet disorders, and other RBDs. The purpose of the WFH report is to provide useful information to hemophilia organizations, hemophilia treatment centers, and health officials involved in efforts to reduce or prevent complications of bleeding disorders and to assist with program planning. The last survey (2007, courtesy of Dr. Mark Brooker) includes data from 98 countries. The RBDD was developed in 2004 by the Angelo Bianchi Bonomi Haemophilia and Thrombosis Centre and the University of Milan. This center has been collecting information and establishing international collaborations through the development of an RBDs network, becoming an international reference point.⁵ The continuing collaboration with different international centers allowed the further extension of knowledge on RBDs and provides a platform to focus and develop further

diagnostic and therapeutic research activity. Since 2005, on the www.rbdd.org Web site, a survey collecting information on type of deficiency, treatment, and related problems has been available. Currently, data has been collected on patients from as many as 68 different countries. The ultimate goal of the RBDD survey is to provide information on the global prevalence of RBDs, along with the availability of treatment facilities in different areas of the world. We decided to compare the data obtained by the two surveys with each other and then also with those reported by six national registries (Central and South America, Egypt, France, India, Iran, United Kingdom). Surveys and registries obtained similar results (Fig. 1), confirming that FVII and FXI deficiencies are the most prevalent RBDs (representing ~28 to 31% of the total number of affected patients), followed by fibrinogen, FV, and FX (7 to 10%), FXIII (~6%), and FV + FVIII deficiencies (3 to 4% from the WFH survey and the six national registries, and 15% from the RBDD survey), and the rarest disorder was found to be FII deficiency with a prevalence of 2%. Thus, results from the surveys/registries differ only with respect to the combined FV + FVIII deficiency, perhaps due to the fact that in the RBDD, cases of FV deficiency associated with mild hemophilia might be misdiagnosed as combined FV + FVIII deficiency.⁵

CLINICAL FEATURES

There are considerable variations in bleeding patterns among patients with RBDs, but on the whole, bleeding symptoms that endanger life, such as central nervous system (CNS) and musculoskeletal bleeding, appear to

Table 1 Bleeding Symptoms in Patients with RBDs

Symptoms	Coagulation Factor Deficiency							
	Fibrinogen	FII	FV	FV + FVIII	FVII	FX	FXI	FXIII
Nosebleeds	Common	Common	Common	Occasional	Common	Common	Common	Common
Cutaneous	Common	NA	Common	Common	Common	Common	Common	Common
Menorrhagia	Common	Common	Common	Common	Common	Occasional	Common	Occasional
Hematuria	Absent	Rare	Absent	Absent	Rare	Occasional	Absent	Occasional
Gastrointestinal bleeding	Occasional	Occasional	Occasional	Absent	Occasional	Common	Occasional	Occasional
Joint bleeding	Common	Common	Rare	Rare	Occasional	Common	Common	Common
Muscle bleeding	Common	Common	Occasional	Occasional	Occasional	Common	Rare	Occasional
Umbilical cord bleeding	Common	Occasional	Absent	Absent	Rare	Common	Absent	Common
Central nervous system bleeding	Occasional	Rare	Rare	Absent	Occasional	Occasional	Absent	Common
Oral cavity bleeding	Common	Common	Common	Common	Common	Common	Occasional	Common
Pregnancy/delivery*	Absent	NA	Absent	Absent	Occasional	Absent [†]	Absent	Absent [†]
Major surgery*, [†]	Occasional	Occasional	Occasional	Common	Occasional	Common	Common	Absent
Minor surgery [†]	Common	Occasional	Occasional	Common	Common	Common	Common	Common
Other	Rare	NA	Rare	Occasional	Absent	Occasional	Rare	Absent
None	Absent	Absent	Absent	Absent	Absent	Absent	Occasional	Absent

*Treatment could not be ruled out.

[†]Percentages were calculated on the basis of the numbers of procedures.

[‡]Percentages were calculated on one patient.

Note: Data are extrapolated by the International Rare Bleeding Disorders Database (RBDD, www.rbdd.org, tailored questionnaire). Data on the inherited deficiency of fibrinogen, FV, FVII, FX, FXI, and FXIII refer to patients with severe factor deficiency (coagulant activity less than 5% and fibrinogen coagulant activity less than 5 mg/dL), whereas data on FV + FVIII refer to patients with moderate deficiency (coagulant activities between 5% and 10%). Pertaining to bleedings during pregnancy/delivery and major surgery, it cannot be ruled out that some patients, particularly patients with FXIII deficiency, were on prophylaxis. *Rare* denotes symptoms present in 0 to 10% of patients; *occasional* denotes symptoms present in 10 to 30% of patients; *common* denotes symptoms present in >30% of patients. NA, data not available.

be less frequent than that in hemophilia.³ Table 1 shows data extrapolated from the RBDD (www.rbdd.org, tailored questionnaire) on the frequency of each bleeding symptom in 117 patients affected by severe RBDs, with the exclusion of 13 patients who had moderate FV + FVIII deficiency. The most typical symptom, common to all RBDs, is the occurrence of excessive bleeding at the time of invasive procedures, delivery in women, and circumcision in men. Mucosal tract bleeding episodes such as epistaxis and menorrhagia were also among the most frequent symptoms. CNS bleeding is a more frequent symptom in FXIII, fibrinogen, FVII, and FX deficiencies than in other RBDs, and umbilical cord bleeding is prevalent in afibrinogenemia, FXIII, and FX deficiency. Gastrointestinal bleeding occurs mainly in FX deficiency but less frequently is also reported in all other RBDs, with the exclusion of combined FV + FVIII deficiency. Limb-endangering hemarthroses are mostly present in severe afibrinogenemia, FII, FX, FXI, and FXIII deficiencies, and soft tissue hematomas occur with the highest frequency in fibrinogen, FII, and FX deficiencies. Detailed information on the clinical picture of each bleeding disorder is reported by each author in the corresponding articles of this issue of *Seminars in Thrombosis and Hemostasis*.

RBDs cause important clinical and social problems for women due to bleeding complications during pregnancy and after delivery but also particularly due to

menorrhagia, as this is reported in ~50% of women with RBDs, with no difference among each bleeding disorder.⁶⁻⁸ Affected women suffer also from the secondary consequences of menorrhagia, such as chronic iron deficiency anemia and a significantly reduced quality of life, due mainly to a complicated reproductive life but also due to limitation in social activities and work.^{9,10} Recurrent miscarriages are frequently described only in afibrinogenemic and FXIII-deficient women (see corresponding articles in this issue of *Seminars in Thrombosis and Hemostasis*). Postpartum bleeding often occurs if replacement therapy is not administered for at least 2 or 3 days after delivery. Surprisingly and paradoxically, thrombotic episodes are reported in patients with fibrinogen and FVII deficiencies; this issue will be discussed in-depth in the specific articles.

DIAGNOSIS

Phenotype Analysis

It is important that each laboratory (i) establishes a local normal range for each performed assay, (ii) assesses the sensitivity of screening assays for respective factor deficiencies, and (iii) that each participates in both internal and external quality assessment schemes.¹¹⁻¹⁸ Prothrombin time (PT) and activated partial thromboplastin time (APTT) are usually sensitive enough to detect FV,

FV + FVIII, FVII, FX, or FXI deficiencies, but they may be normal in fibrinogen, FII, and FXIII deficiencies, which therefore require specific assays. The thrombin time (TT) is required for the diagnosis of fibrinogen deficiency in addition to the PT and APTT. The diagnosis of dysfibrinogenemia requires also an immunoreactive fibrinogen assay in addition to coagulation functional assay, as the disease is expressed by a discrepancy between clottable and immunoreactive fibrinogen. For the same reason, an immunologic assay is essential for the diagnosis of dysprothrombinemia, and without obtaining prothrombin antigen levels, patients with prothrombin deficiency cannot be properly classified. FXIII deficiency was originally diagnosed by observing the solubility of the fibrin clot in concentrated urea or monochloroacetic acid solutions. However, this method detects only very severe FXIII deficiency (with activity below 0.5 to 3%), so that it carries a risk of underdiagnosing this deficiency. The appropriate laboratory assays for identification of each deficiency will be described in detail in the articles within this issue of *Seminars in Thrombosis and Hemostasis*.

Genotype Analysis

The molecular diagnosis of RBDs is based on the mutation search in the genes encoding the corresponding coagulation factors. Exceptions are the combined deficiency of coagulation FV + FVIII, caused by mutations in genes encoding proteins involved in the FV and FVIII intracellular transport (MCFD2 and LMAN1) and the combined deficiency of vitamin K-dependent proteins (FII, FVII, FIX, and FX), caused by mutations in genes that encode enzymes involved in posttranslational modifications¹⁹ and in vitamin K metabolism (GGCX and VKORC1).²⁰ The detailed and most up-to-date data on gene mutation analysis for each gene encoding the corresponding protein are described in each article of this issue of *Seminars in Thrombosis and Hemostasis*. The broad analysis of genetic data contained in the RBDD on 357 patients (www.rbdd.org) and compared with data published and updated by Zivelin et al on the International Society on Thrombosis and Haemostasis (ISTH) Web site (<http://www.isth.org/publications/registriesdatabases/mutationsrarebleeding-disorders/asp>), show a similar distribution of mutation types. Missense mutations are the most prevalent, being present in 52 to 57% of patients with different RBDs, followed by small insertion/deletion mutations in 17 to 20% of patients, nonsense mutations in 10 to 13%, and splicing site mutations in 13%. The next step will be to collectively analyze the data on gene analysis obtained by national, international, and/or different human genetic databases, which in the past decade have relevantly increased the amount of available data on each specific RBD. These data will be important to identify common

and recurrent gene mutations to allow the design of algorithms for mutation detection in new cases for each disorder in each specific region of the world. Recurrent mutations are respectively reported for each RBD in the accompanying articles of this issue. For instance, the IVS4 + 1 mutation on the *FGA* gene seems to be the first mutation that should be looked for in European patients with afibrinogenemia, whereas the Gln117stop and Phe283Leu mutations on the *FXI* gene account for 95% of patients with FXI deficiency among Ashkenazi Jews. In addition, the Arg77His mutation on the *FXIII* gene is present in 33% of Iranian patients with severe FXIII deficiency.^{21,22} These and other genetic variants could be markers for quick and convenient genotyping, particularly for prenatal diagnosis in those families with one affected child already presenting with severe bleeding symptoms, particularly in countries with low economic resources where genetic prevention could improve the quality of life of these families. Genetic analysis is also important for preimplantation genetic diagnosis (PGD),²³ especially in those countries with a high rate of RBDs and consanguinity, particularly where genetic prevention by premature termination of pregnancy is not allowed. As mutation detection is a relatively easy and available strategy in different laboratories worldwide, and it is not always clear whether or not the identified mutation could be the cause of the severe deficiency of a specific clotting factor, family history associated with allele frequency in the genetic population is important to exclude the possibility that identified mutations are common polymorphisms. Finally, in vitro expression studies of each gene mutation, together with biochemical characterization of the mutant protein, could be important to establish the mechanistic link between gene mutations and the observed disease phenotypes.²⁴

TREATMENT

Treatment of RBDs is a difficult task, as information on the clinical management of RBDs is often scarce, and replacement therapy of coagulation factors may require the prescription of unlicensed products that are not readily available.² The patients' personal and family histories of bleeding are important guides for management.² Dosages and frequency of treatment (described in each article of this issue for each single deficiency) depend on the minimal hemostatic level of the deficient factor, its plasma half-life, and the type of bleeding episode.²

The main treatments in RBDs are the following^{3,4}:

1. Non-transfusional adjuvant therapies: Antifibrinolytic amino acids can be useful, alone or in combination with replacement therapy, in the management of

the less severe forms of mucosal tract hemorrhages. Epsilon-aminocaproic acid and tranexamic acid should be administered orally or intravenously. The continued use of estrogen-progestin preparations helps to reduce menstrual blood loss causing iron deficiency and chronic anemia in women with menorrhagia.²⁵

2. Replacement therapy: As for hemophilia, replacement of the deficient coagulation factor is the mainstay of treatment for RBDs. Unfortunately, information on the safety and efficacy of the few available products is scarce, and experience with their optimal use is much more limited than for those available for hemophilia. Data on heterozygous patients undergoing surgery and on severe patients on prophylaxis are very scarce.^{2,4}

Fresh-frozen plasma (FFP) and cryoprecipitate are the backbone of RBDs treatment everywhere. The recent survey of 68 treatment centers variably distributed throughout the world (<http://www.rbdd.org/fig9.htm>) showed that more than 50% of patients are still treated only with FFP because this product is relatively inexpensive and widely available. However, infectious complications with such blood-borne viruses as the hepatitis viruses or the human immunodeficiency virus (HIV) are still perceived as a threat associated with the use of FFP.³ Therefore, even though improvements in donor selection and screening, including nucleic acid testing, is likely to have minimized the actual absolute risk of blood-borne infections, virus-inactivated FFP is preferable to plain FFP.²⁶ Unfortunately, most countries have no access to virus-inactivated FFP. Plasma-derived concentrates of fibrinogen, FVII, FXI, and FXIII are currently available, but they are licensed only in some European countries. Prothrombin and FX deficiencies are often treated with prothrombin complex concentrates (PCCs) that, though licensed for treatment of hemophilia B, contain also large and often uncontrolled amounts of FII, FVII, and FX. Therefore, patients should be closely monitored to avoid the possibility that each factor rises in excess of 150%, with an associated increased risk of thrombosis. A recently developed product by CSL Behring (Factor X P Behring, King of Prussia, PA), containing a known amount of FX but also of FIX, could be helpful to tackle this problem, before specific FII and FX plasma-derived concentrates are developed, particularly in patients who require regular prophylaxis. Only one recombinant product is available for treatment of RBDs, namely recombinant FVIIa, licensed for the treatment of FVII deficiency.²⁷ Because this product is active as a general hemostatic agent that potentiates coagulability, it has been used in a few cases of FV and FXI deficiencies.^{28–32} However, more accurate data are required to establish the true usefulness of this product in the latter type of RBDs. Unfortunately,

there is no FV concentrate available, and patients with FV and combined FV + FVIII deficiencies can be treated only with FFP, preferentially virus inactivated. However, fluid overload is a potential problem, particularly when repeated infusions are required, and plasma exchange may be necessary in these cases.⁴ In conclusion, replacement therapy is an effective treatment of the bleeding episodes, but due to the lack of accurate pharmacokinetics studies, the effectiveness of treatment could be highly variable among patients, and it is important to individually adjust treatment, particularly in children.

Recommendations for primary prophylaxis are made difficult by the poor availability of data, except for FXIII deficiency.⁴ Secondary prophylaxis might be appropriate in cases with potential life-threatening hemorrhages such as CNS bleeding because of the risk of recurrences, particularly in FXIII, FX, and FVII deficiencies. This approach should also be considered in some clinically severe cases of afibrinogenemia and FV deficiency.⁴

The best management of pregnancy should be decided through the coordination of a team composed of pediatricians, hematologists, and obstetricians. In addition to female health care, the same team should ensure investigation and management of potentially affected newborns, particularly in families already containing one affected child. Severe FVII, FX, and FXIII deficiencies present with a particularly high risk of intracranial hemorrhage during the first week of life.

CONCLUSIONS

Even though in the past decade substantially more knowledge on the prevalence, clinical manifestation, and genotype analysis of RBDs has been achieved, additional efforts are required to ameliorate the global care of RBDs:

The definition of the minimal hemostatic level of each coagulation factor is still an unresolved problem. Global hemostasis assays need to be developed to better evaluate the hemostasis potential of each patient, because it is well known from clinical experience that, in patients with RBDs, the relation between prolongation of coagulation screening tests, plasma levels of deficient factors, and bleeding tendency can vary markedly from deficiency to deficiency and patient to patient. Hence, the conventional functional assays for measurement of factor levels (based on the PT, APTT, and TT) are of limited help in clinical practice and relatively insensitive for mild coagulation abnormalities. The actual available assays lack the ability to screen an increased thrombophilic signature that can partially compensate for the coagulation factor defect. Perhaps a global test such as thrombin generation might be found to be better

related to the severity of bleeding symptoms in RBDs.^{33,34}

There is still relatively little information on the relation between laboratory phenotype and the clinical pattern of symptoms. Therefore, a standardized quantitative assessment of bleeding needs to be developed for the evaluation of its severity and the definition of what can be considered a significant bleeding tendency. This approach will help to distinguish patients with a borderline history, affected with partial coagulation deficiency (heterozygotes) and who may require a specific treatment only during surgery, from those with severe bleedings and severe factor deficiency who might require prophylactic treatment. The development of a bleeding severity score could be useful in the choice of the optimal treatment.

Last, but not least, particular attention is warranted on the health care status of women affected by RBDs. It is necessary to conduct appropriately designed studies for prompt diagnosis, prophylaxis, or treatment that may significantly improve the usually poor quality of life of these women.

REFERENCES

- Tuddenham EGD, Cooper DN. The Molecular Genetics of Haemostasis and Its Inherited Disorders. Oxford, UK: Oxford Medical Publications; 1994
- Peyvandi F, Kaufman RJ, Seligsohn U, et al. Rare bleeding disorders. *Haemophilia* 2006;12(Suppl 3):137–142
- Mannucci PM, Duga S, Peyvandi F. Recessively inherited coagulation disorders. *Blood* 2004;104:1243–1252
- Bolton-Maggs PH, Perry DJ, Chalmers EA, et al. The rare coagulation disorders—review with guidelines for management from the United Kingdom Haemophilia Centre Doctors' Organisation. *Haemophilia* 2004;10:593–628
- Peyvandi F, Cattaneo M, Inbal A, De Moerloose P, Spreafico M. Rare bleeding disorders. *Haemophilia* 2008;14(Suppl 3):202–210
- Greer IA, Lowe GD, Walker JJ, Forbes CD. Haemorrhagic problems in obstetrics and gynaecology in patients with congenital coagulopathies. *Br J Obstet Gynaecol* 1991;98:909–918
- Lee CA, Chi C, Pavord SR, et al; UK Haemophilia Centre Doctors' Organization. The obstetric and gynaecological management of women with inherited bleeding disorders—review with guidelines produced by a taskforce of UK Haemophilia Centre Doctors' Organization. *Haemophilia* 2006;12:301–336
- Kumar M, Mehta P. Congenital coagulopathies and pregnancy: report of four pregnancies in a factor X-deficient woman. *Am J Hematol* 1994;46:241–244
- Kadir RA, Sabin CA, Pollard D, Lee CA, Economides DL. Quality of life during menstruation in patients with inherited bleeding disorders. *Haemophilia* 1998;4:836–841
- Kouides PA, Phatak PD, Burkart P, et al. Gynaecological and obstetrical morbidity in women with type I von Willebrand disease: results of a patient survey. *Haemophilia* 2000;6:643–648
- Lindley C, Sawyer W, Haddon T, et al. Comparison of PT, aPTT, and factor VII values obtained by concurrent sample collection by direct venipuncture and peripheral venous catheters. *Pharmacotherapy* 1994;14:224–228
- Almadrones L, Godbold J, Raaf J, Ennis J. Accuracy of activated partial thromboplastin time drawn through central venous catheters. *Oncol Nurs Forum* 1987;14:15–18
- Adcock DM, Hoefner DM, Kottke-Marchant K, Marlar RA, Szamosi DI, Warunek DJ. Collection, Transport, and Processing of Blood Specimens for Testing Plasma-Based Coagulation Assays and Molecular Hemostasis Assays: Approved Guideline—Fifth Edition. Wayne, PA: Clinical Laboratory Standards Institute; 2008; CLSI document H21–A5
- WHO. Use of Anticoagulants in Diagnostic Laboratory Investigations. Geneva, Switzerland: WHO; 1999:1. WHO/DIL/LAB/99.1
- CISMEL. Activated partial thromboplastin time: a multi-centre evaluation of 11 reagents in the screening of mild haemophilia A. *Scand J Haematol* 1980;25:308–317
- Hoffmann JJ, Meulendijk PN. Comparison of reagents for determining the activated partial thromboplastin time. *Thromb Haemost* 1978;39:640–645
- Lawrie AS, Kitchen S, Purdy G, Mackie IJ, Preston FE, Machin SJ. Assessment of Actin FS and Actin FSL sensitivity to specific clotting factor deficiencies. *Clin Lab Haematol* 1998;20:179–186
- Turi DC, Peerschke EI. Sensitivity of three activated partial thromboplastin time reagents to coagulation factor deficiencies. *Am J Clin Pathol* 1986;85:43–49
- Zhang B, Cunningham MA, Nichols WC, et al. Bleeding due to disruption of a cargo-specific ER-to-Golgi transport complex. *Nat Genet* 2003;34:220–225
- Sadler JE. Medicine: K is for koagulation. *Nature* 2004;427:493–494
- Peyvandi F, Tagliabue L, Menegatti M, et al. Phenotype-genotype characterization of 10 families with severe a subunit factor XIII deficiency. *Hum Mutat* 2004;23:98
- Cairo A, Menegatti M, Spreafico M, Garagiola I, Peyvandi F. Recurrent mutations identified on LMAN1 and coagulation factors VII, X, XIII genes [abstract]. *Haematologica* 2008;93(s3):97
- Handyside AH, Kontogianni EH, Hardy K, Winston RM. Pregnancies from biopsied human preimplantation embryos sexed by Y-specific DNA amplification. *Nature* 1990;344:768–770
- Peyvandi F, Jayandharan G, Chandy M, et al. Genetic diagnosis of haemophilia and other inherited bleeding disorders. *Haemophilia* 2006;12(Suppl 3):82–89
- Kwiecien M, Edelman A, Nichols MD, Jensen JT. Bleeding patterns and patient acceptability of standard or continuous dosing regimens of a low-dose oral contraceptive: a randomized trial. *Contraception* 2003;67:9–13
- Santagostino E, Mancuso ME, Morfini M, et al. Solvent/detergent plasma for prevention of bleeding in recessively inherited coagulation disorders: dosing, pharmacokinetics and clinical efficacy. *Haematologica* 2006;91:634–639
- Brenner B, Wiis J. Experience with recombinant-activated factor VII in 30 patients with congenital factor VII deficiency. *Hematology* 2007;12:55–62
- González-Boullousa R, Ocampo-Martínez R, Alarcón-Martín MJ, Suárez-Rodríguez M, Domínguez-Viguera L, González-Fajo G. The use of activated recombinant coagulation factor VII during haemarthroses and synovectomy in a

- patient with congenital severe factor V deficiency. *Haemophilia* 2005;11:167–170
29. Divanon F, Hecquard C, Borel-Derlon A. Experience with use of recombinant activated factor VII. *J Clin Pharm Ther* 2002;27:133–138
 30. Altisent C, Lozano M, Sol E, et al. Dental extraction using rFVIIa in a patient with severe FV deficiency. *Haemophilia* 2000;6:408
 31. O'Connell NM. Factor XI deficiency—from molecular genetics to clinical management. *Blood Coagul Fibrinolysis* 2003;14(Suppl 1):S59–S64
 32. O'Connell NM, Riddell AF, Pascoe G, Perry DJ, Lee CA. Recombinant factor VIIa to prevent surgical bleeding in factor XI deficiency. *Haemophilia* 2008;14:775–781
 33. Al Dieri R, Peyvandi F, Santagostino E, et al. The thrombogram in rare inherited coagulation disorders: its relation to clinical bleeding. *Thromb Haemost* 2002;88:576–582
 34. Hemker HC, Giesen P, AlDieri R, et al. The calibrated automated thrombogram (CAT): a universal routine test for hyper- and hypocoagulability. *Pathophysiol Haemost Thromb* 2002;32:249–253